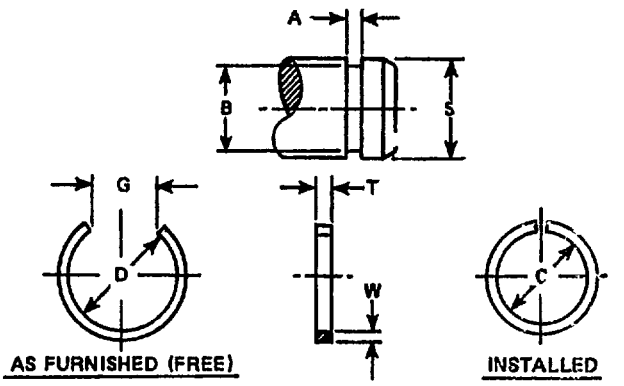
USER ACTIVITIES: WC

This military standard is approved by the Naval Sea Systems Command, Department of the Navy and shall be used by that activity. All other military activities are required to employ this standard where suitable.



DASH NO.	SHAFT DIAMETER		RECOMMENDED GROOVE DIMENSIONS				RING DIMENSIONS								
							As Furnished (Free)								Installed
			Diameter		Width		I.D.		Gap		Width		Thickness		I.D.
See Note 4	Nominal S	Basic S	B		A		D		G		W		T		C
1	3/16	0.188	.125	+.000 −.005	.034	+.003 −.000	0.197	+.000 −.005	.125	+.005 −.000	.062	±.015	.031	±.002	.125
2	7/32	0.219	.173		.034		0.230	+.000 −.010	.143		.046		.031		.173
3	1/4	0.250	.188		.034		0.266		.182		.062		.031		.188
4	5/16	0.312	.250		.043		0.328		.191		.079		.039		.250
5	3/8	0.375	.313	+.000 −.010	.052	+.005 −.000	0.394		.264	+.010 −.000	.094		.047	±.003	.313
6	7/16	0.438	.376		.060		0.459	+.000 −.015	.203		.109		055		.376
7	1/2	0.500	.438		.069		0.525		.233		.125		.063		.438
8	9/16	0.562	.493		.077		0.591	+.000 −.020	.241		.141		.070		493
9	5/8	0.625	.547		.086		0.656		.253		.156		.078	±.001	547
10	11/16	0.688	.602		.094		0.722		.278		.172		.086		.602
11	3/4	0.750	.656		.103		0.788	+.000 −.025	.310		.188		094		656
12	13/16	0.812	711		.112		0.853		.331	+.015 −.000	.203		.102	±.005	.711
13	7/8	0.875	.765		.120		0.919	+.000 −.030	.357		.219		.109		.765
14	15/16	0.938	.820		.129		0.984		.387		.234		117		.820
15	1	1.000	.874		.138		1.050		.423		.250		.125		874

For NOTES see SHEET 2.

Ⓐ ENTIRE STANDARD REVISED

P A OS Other Cust	INTERNATIONAL INTEREST	TITLE: RING, RETAINING, EXTERNAL	MILITARY STANDARD MS3211(OS)
PROCUREMENT SPECIFICATION: None	SUPERSEDES: NAVSEA (Code Ident. 10001) Dwg 2533400		SHEET 1 OF 2

APPROVED 24 April 1967 REVISED Ⓐ 21 March 1970

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FED. SUP CLASS
5365

NOTES:

1. MATERIAL:

(a) Carbon steel in accordance with FED-STD-66, AISI 1008 thru 1020 inclusive.

(b) Corrosion-resistant steel (CRES) in accordance with QQ-S-763, Class 302 or 304, annealed; QQ-S-766, Class 302 or 304, strip, annealed; or ASTM A 167, Type 302 or 304, strip, annealed.

(c) Phosphor bronze in accordance with ASTM B 159, copper alloy No 510, soft; ASTM B 103, copper alloy No. 510 strip, soft, or QQ-B-750, Composition A, strip or flat wire, soft

(d) Beryllium copper, copper alloy No. 172, in accordance with ASTM B 197, annealed; ASTM B 194 strip, annealed; or QQ-C-533 strip, Temper A

2 PROTECTIVE COATING:

(a) Carbon steel rings shall be zinc plated in accordance with QQ-Z-325, Type II, Class 2 or MIL-C-81562, Class 2, Type II

(b) CRES rings shall be passivated in accordance with QQ-P-35.

(c) Beryllium copper rings requiring protective coating shall be cadmium plated in accordance with QQ-P-416, Class 2, Type II or MIL-C-81562, Class 2, Type II.

3 TOLERANCE: Tolerances apply before application of protective coating.

4 PART NUMBER. The MS part number consists of the MS number, plus the dash number, plus the material designator

MATERIAL DESIGNATOR	USE AFTER DASH NO. TO SPECIFY:
-SZ	Zinc Plated Carbon Steel Ring
-CR	Corrosion-Resistant Steel Ring
-PB	Phosphor Bronze Ring
-BX	Beryllium Copper Ring
-BC	Cadmium Plated Beryllium Copper Ring

Example: MS3211-1-SZ is the part number for a zinc plated carbon steel external retaining ring with as furnished I.D of 0.197 and installed I.D. of 0 125 for use on a 0 188 diameter shaft having a groove diameter of 0.125 and a groove width of 0.034

5 DIMENSIONS: All dimensions are in inches.

6. OTHER NOTES:

(a) Referenced documents shall be of the issue in effect on the date of invitation for bids

(b) For design feature purposes, this standard takes precedence over procurement documents referenced herein

APPROVED 24 April 1967 REVISED Ⓐ For changes see Sheet 1,2

Ⓐ ENTIRE STANDARD REVISED

P.A. OS	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
Other Cust		RING, RETAINING, EXTERNAL	MS3211(OS)
Procurement Specifications None	SUPERSEDES: NAVSEA (Code Ident 10001) Dwg 2533400		SHEET 2 OF 2

DD FORM 1 MAY 73 672 (LIMITED COORDINATION) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5365-N002

USER ACTIVITIES: WC

This military standard is approved by the Naval Sea Systems Command, Department of the Navy and shall be used by that activity. All other military activities are required to employ this standard where suitable.

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